ML GROUP PROJECT

Heart Disease Prediction

Baseline · Improvement · Error Analysis · LLM Feature Engineering

MEMBERS: Huỳnh Gia An · Phạm Đức Anh · Cao Chí Bảo · Nguyễn Tân Nhật Minh

Kaggle Heart Failure Prediction Dataset

Logistic Regression

LLaMA 3.1 via Groq

Project Sources

Project Report

View on GitHub

Kaggle Dataset

View on Kaggle

Project Repository

View on GitHub

Why Does This Matter?

- 17.9 million deaths/year from CVDs — WHO
- Clinical diagnosis is expensive and inaccessible
- Early detection can significantly reduce mortality
- Routine clinical data is enough for ML prediction

→ A preventable condition, if caught early.

17.9M

deaths per year

Cardiovascular disease — #1 cause of death globally

What We Built

01 🔍 **EDA & Preprocessing**

Understand and clean the clinical data

↓

02 🤖 **Logistic Regression**

Build, evaluate, and tune the baseline model

↓

03 🔬 **Error Analysis**

Diagnose model failures and identifying failures

↓

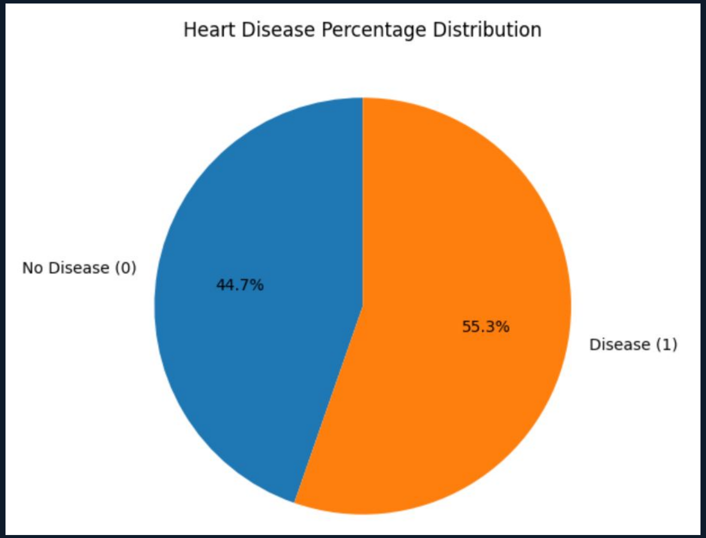
04 🔍 **LLM - Feature Engineering Helper**

Help to suggest feature to improve model

Data Overview

01. Numerical Features

Feature	Description	Unit	Notes
Age	Patient age	years	Range: 28–77
RestingBP	Resting blood pressure	mm Hg	0 values likely data errors
Cholesterol	Serum cholesterol	mg/dL	0 values = missing/imputed
MaxHR	Maximum heart rate achieved	bpm	Higher = better cardiac function
Oldpeak	ST depression (exercise vs rest)	numeric	Higher = more ischemia



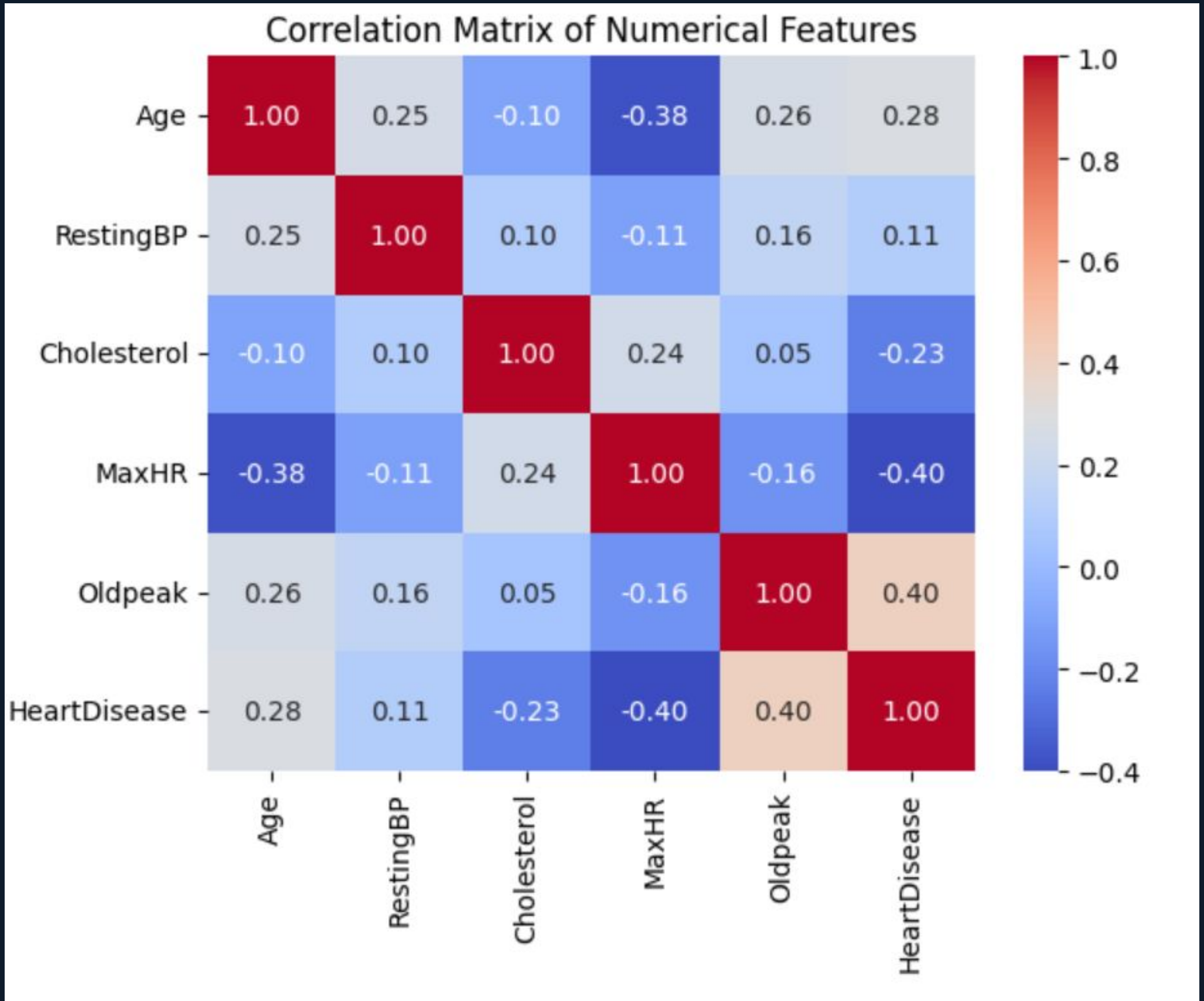
02. Categorical Features

Feature	Description	Values
Sex	Biological sex	M / F
ChestPainType	Type of chest pain	ATA, NAP, ASY, TA
FastingBS	Fasting blood sugar > 120 mg/dL	0 (No) / 1 (Yes)
RestingECG	Resting ECG result	Normal, ST, LVH
ExerciseAngina	Exercise-induced angina	Y / N
ST_Slope	Slope of peak exercise ST segment	Up, Flat, Down

918
Total Patients

55%
Positive Cases

What the Data Tells Us



Key Relationships

MaxHR (−0.40) and Oldpeak (+0.40) show the strongest linear ties to heart disease, marking them as primary predictors.

Moderate Drivers

Age (+0.28) and Cholesterol (−0.23) display notable moderate correlations.

Minimal Impact

Resting BP (+0.11) exhibits the weakest linear correlation among features.

Statistical Integrity

No pairs exceed ±0.70 correlation; multicollinearity is not a concern for Logistic Regression.

PREPROCESSING

Remove Outliers
Resting BP=0
(1 record)

→

Encode
Label + OHE
drop_first=True

→

Split
80/20 Split
Stratified

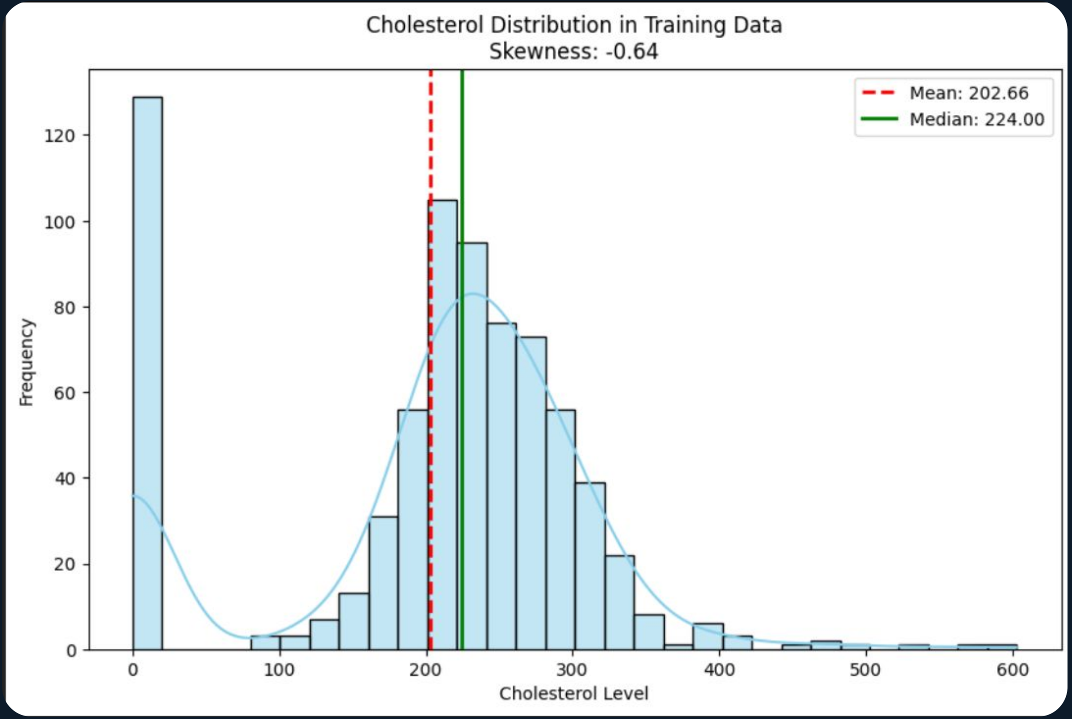
→

Impute
Fit median on Train → Apply on both

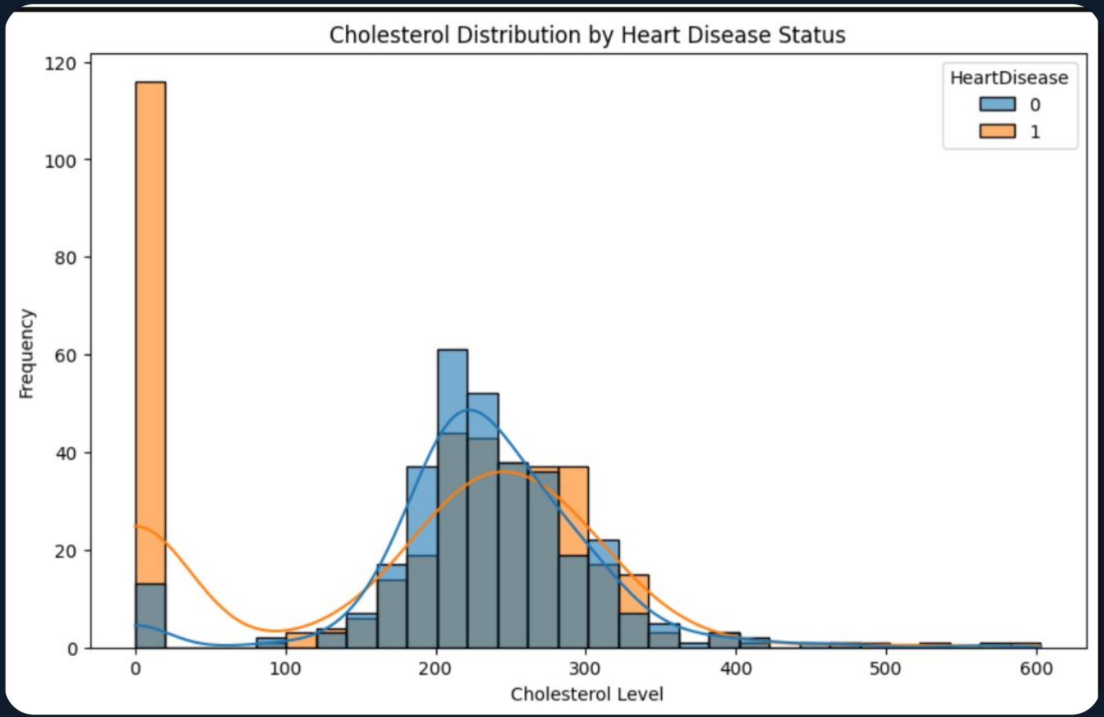
→

Scale
StandardScaler
TRAIN only
⚠ NO LEAKAGE

1. Why encode before splitting?



2. Why use median?



MODEL · BASELINE

Logistic Regression — Baseline

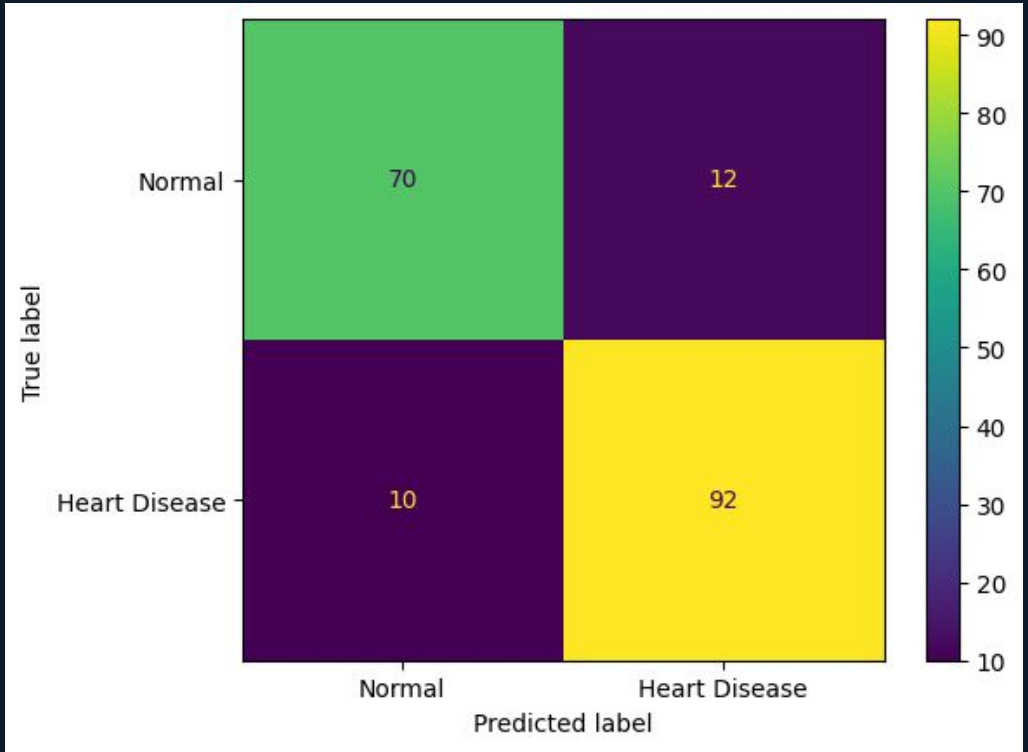
C = 1.0 (default L2)

Threshold = 0.5

Train: 733 | Test: 184

> *"Simple and interpretable — ideal for isolating feature effects"*

⚠ 10% of sick patients are still missed



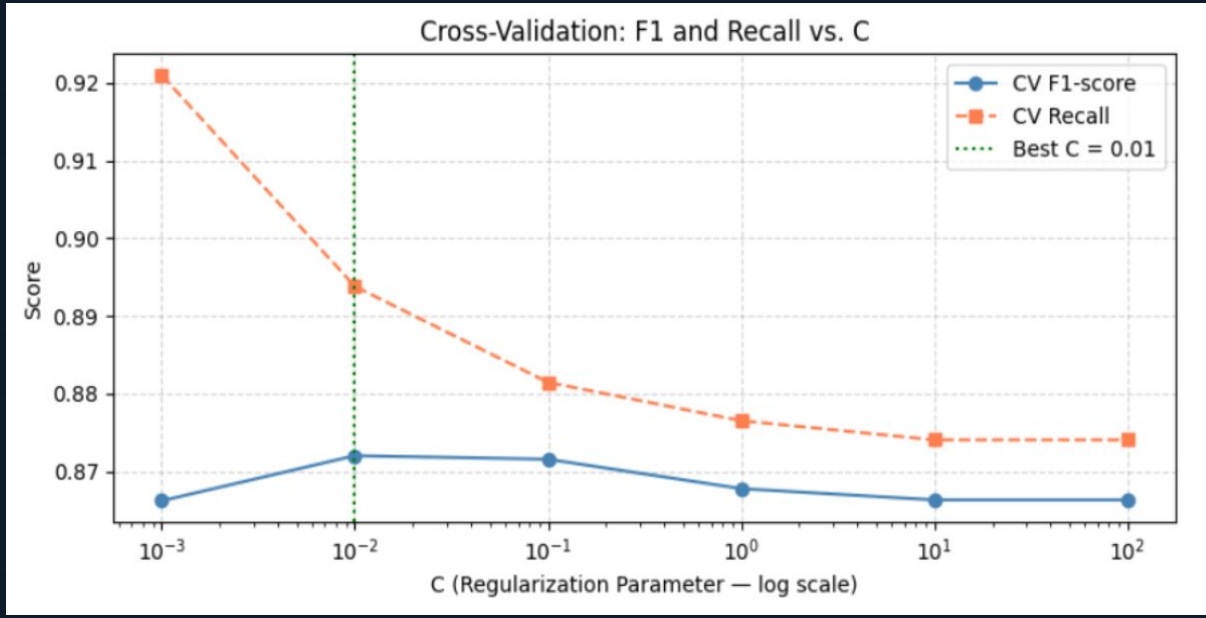
Acc 89.1% | Prec 88.5% | Recall 90.2% | F1 89.3%

* For Heart Disease

MODEL · IMPROVEMENT

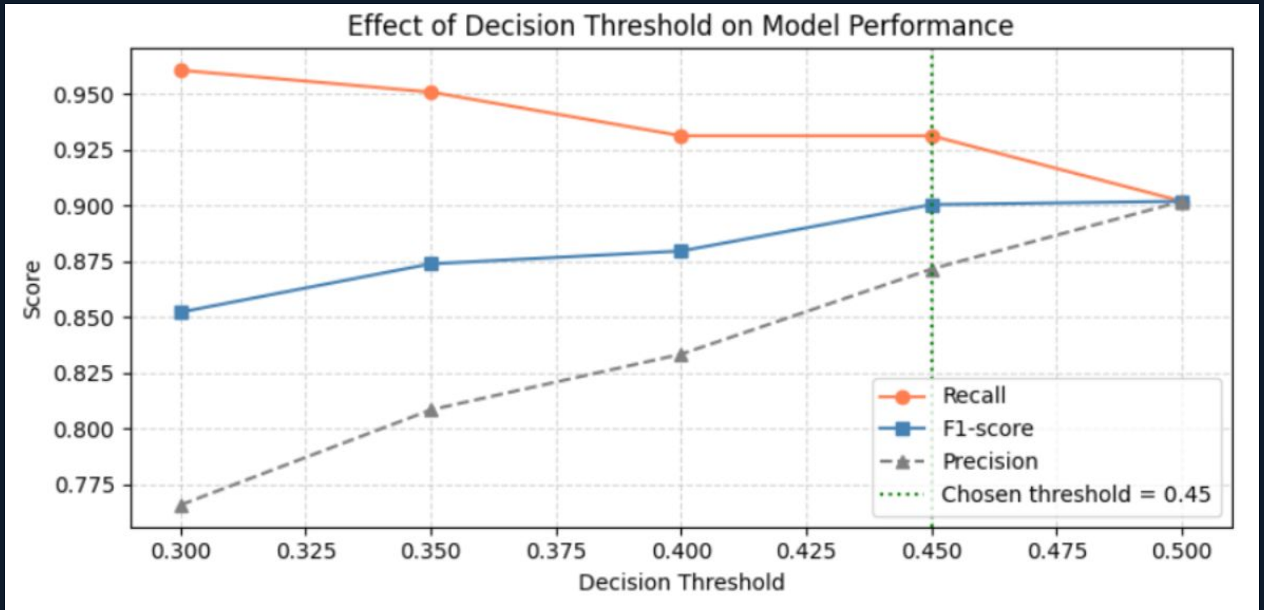
Regularization & Threshold Tuning

C = 0.01



Strongest regularization before underfitting

T = 0.45



Recall ↑ +3.3% · F1 drop < 0.2%

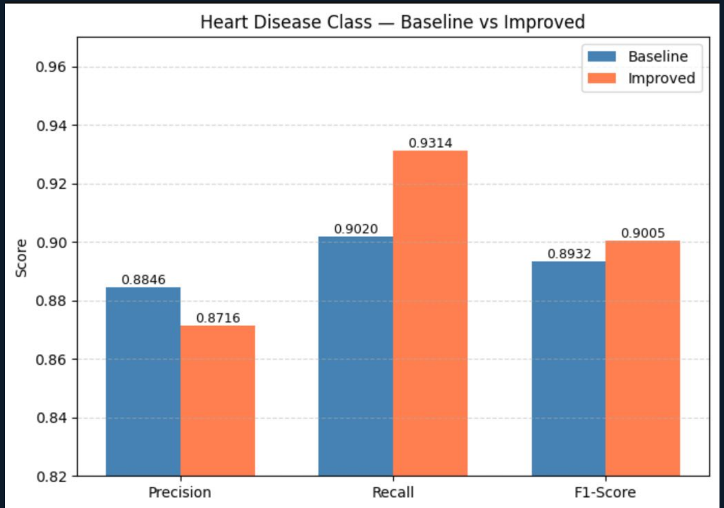
In medicine — missing a sick patient (FN) costs more than a false alarm (FP)

MODEL · RESULTS

The Improvement

*For Heart Disease class

Metric	Baseline	Improved	Δ
Accuracy	89.1%	88.6%	-0.5%
Precision (HD)	88.5%	87.2%	-1.3%
Recall (HD)	90.2%	93.1%	+2.9% ✅
F1-Score (HD)	89.3%	90.1%	+0.7%



+2.9%

Recall improvement

3 more sick patients identified per 100 tested.

ERROR ANALYSIS · 01

When the Model is Wrong

🔴 FP · prob 0.898

Case 174

Key: ExAngina=1, Oldpeak=1.0

"Overconfidence — risk signals dominated protective ones."

🟠 FN · prob 0.273

Case 80

Key: Resting BP=160, Age=59

"High BP underweighted relative to ST_Slope signals."

🟠 FN · prob 0.311

Case 115

Key: Max HR=168, ExAngina=0

"High Max HR masked true disease — silent ischemia."

ERROR ANALYSIS · 02

What Errors Reveal

No feature interaction

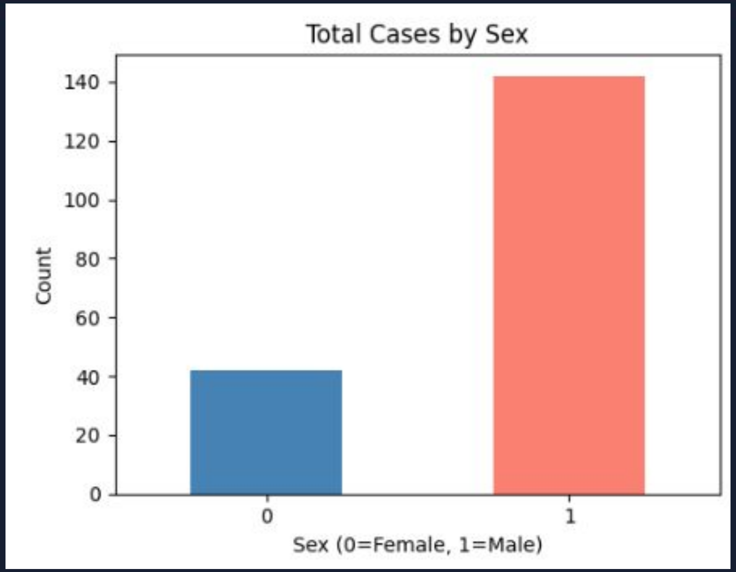
LR adds signals but cannot contextualize conflicting indicators.

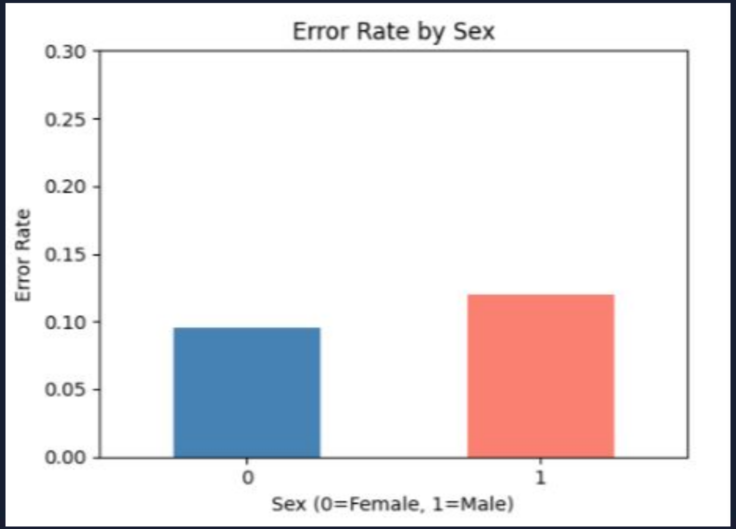
Linear boundary

Cases near the boundary are inherently ambiguous in feature space.

Missing clinical data

BMI, smoking, family history are absent — ceiling effect.





More male cases in training → less reliable for females

LLM COMPONENT · 01

🤖Asking AI to Improve the Model

Model: LLaMA 3.1 (8B)

Prompt: 15 features + clinical context + Z-score constraints

> "We evaluate each suggestion mathematically, not just clinically."

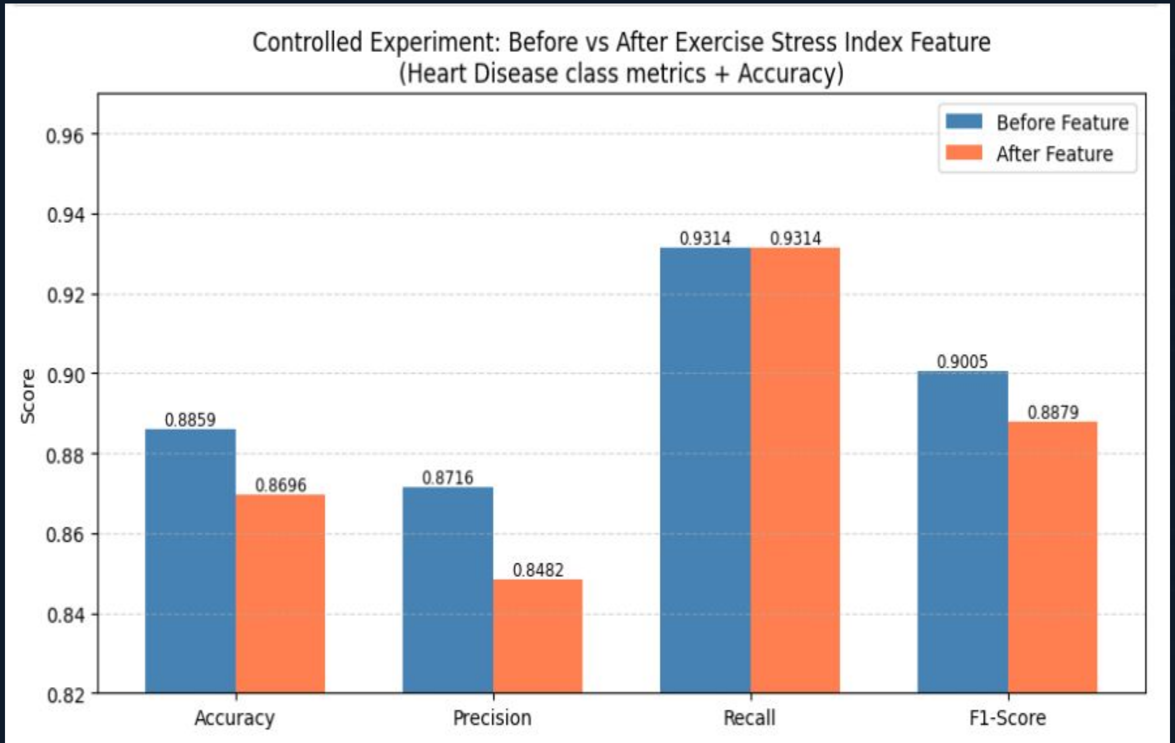
Age × Risk Score ✕
Z-score sign flip creates false directional meaning.

log(Resting BP / Chol) ✕
Standardized denominator can be zero or negative.

Exercise Stress Index ✓
Oldpeak + (Angina × ST_Slope_Up). Clinically sound.

LLM COMPONENT · 02

The Feature Did Not Help



Performance dipped across almost all metrics

Metric	Δ
Accuracy	-1.63%
Precision	-2.34%
Recall	0.00%
F1-Score	-1.26%

⚠ **WHY IT FAILED**

Math distortion: Z-score × Z-score breaks linearity.

Redundancy: LR weights existing features optimally.

CONCLUSION

What We Learned

✅ **LR Tuning:** 88.6% Acc & **93.1% Recall** reached.

✅ **Threshold > Complexity:** Tuning 0.45 threshold yielded the best results.

⚠️ **LLM Bias:** AI excels at domain, fails at math context.

🔬 **Error Ceiling:** Analysis reveals limits of a linear model.

"Clinical validity does not guarantee computational effectiveness."

Future Work:

- 🌲 Try XGBoost
- 📊 Richer Features
- 🧠 Pipeline Prompts